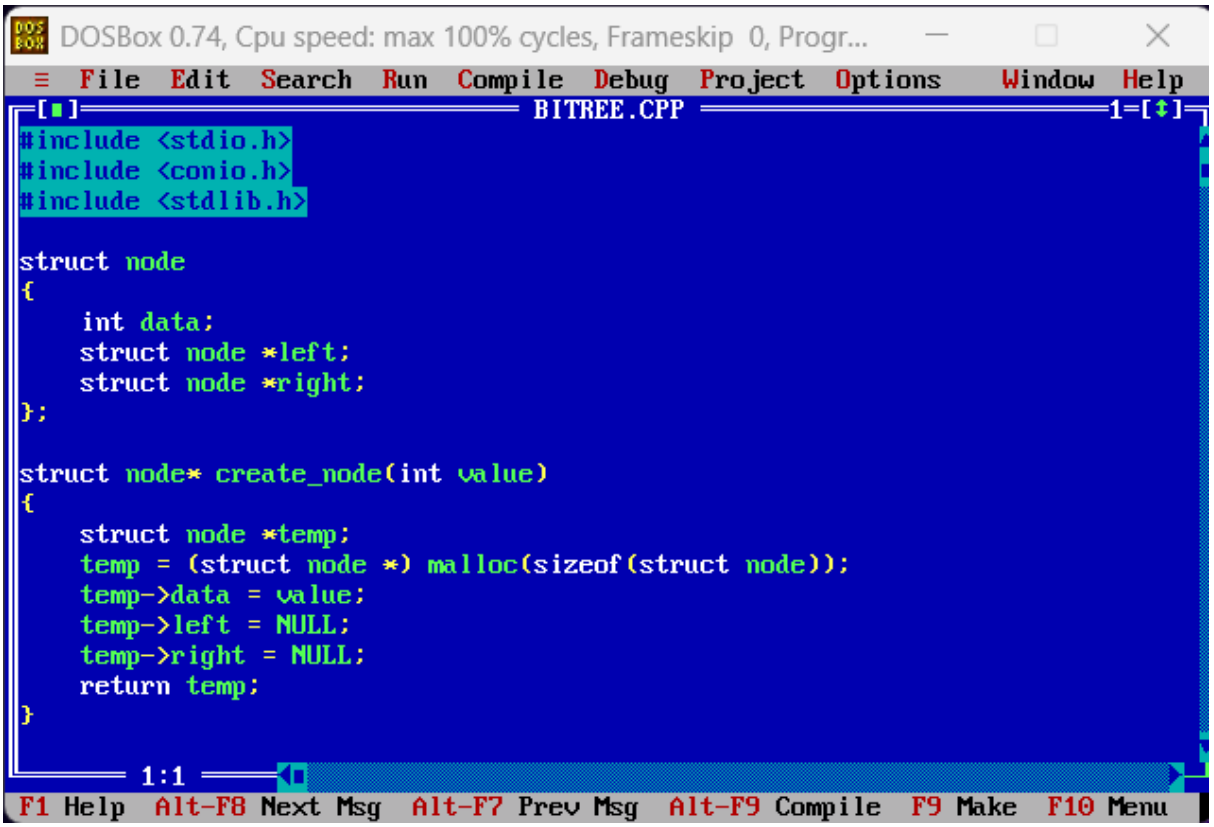
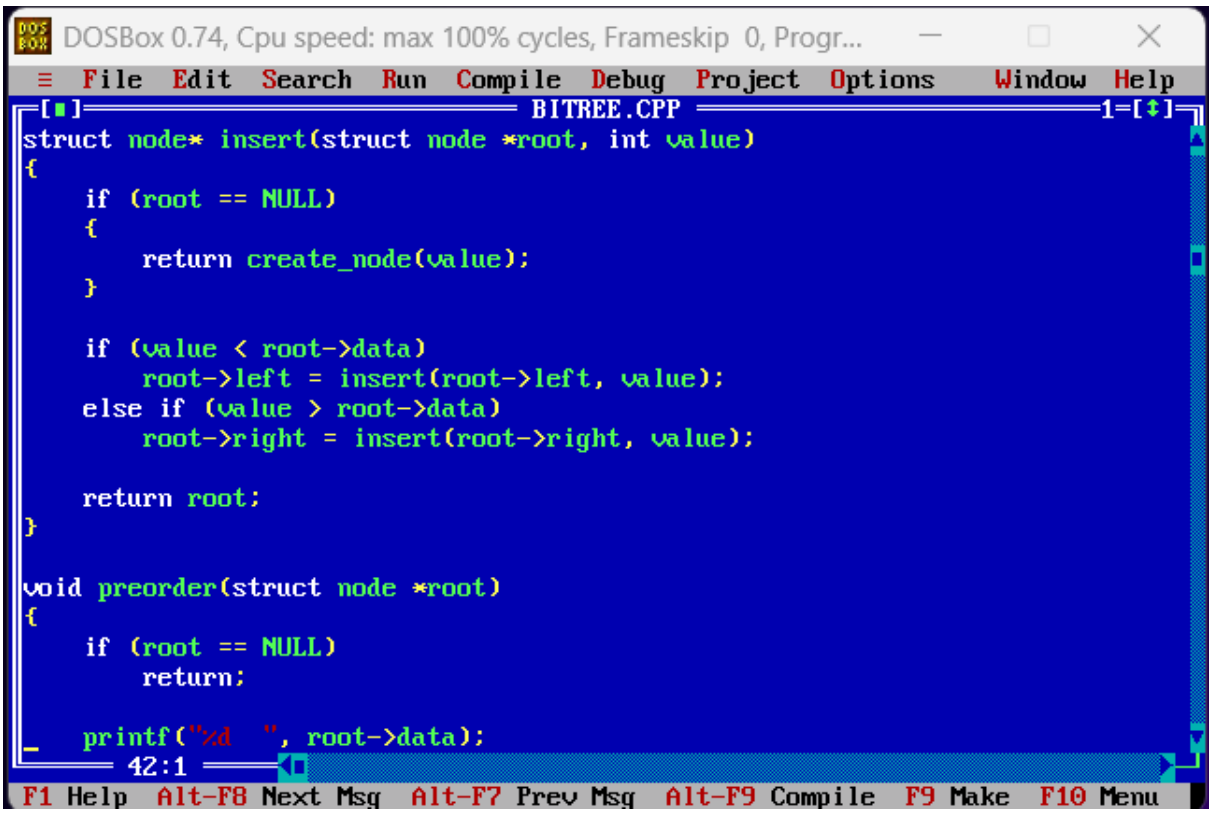
Experiment 6

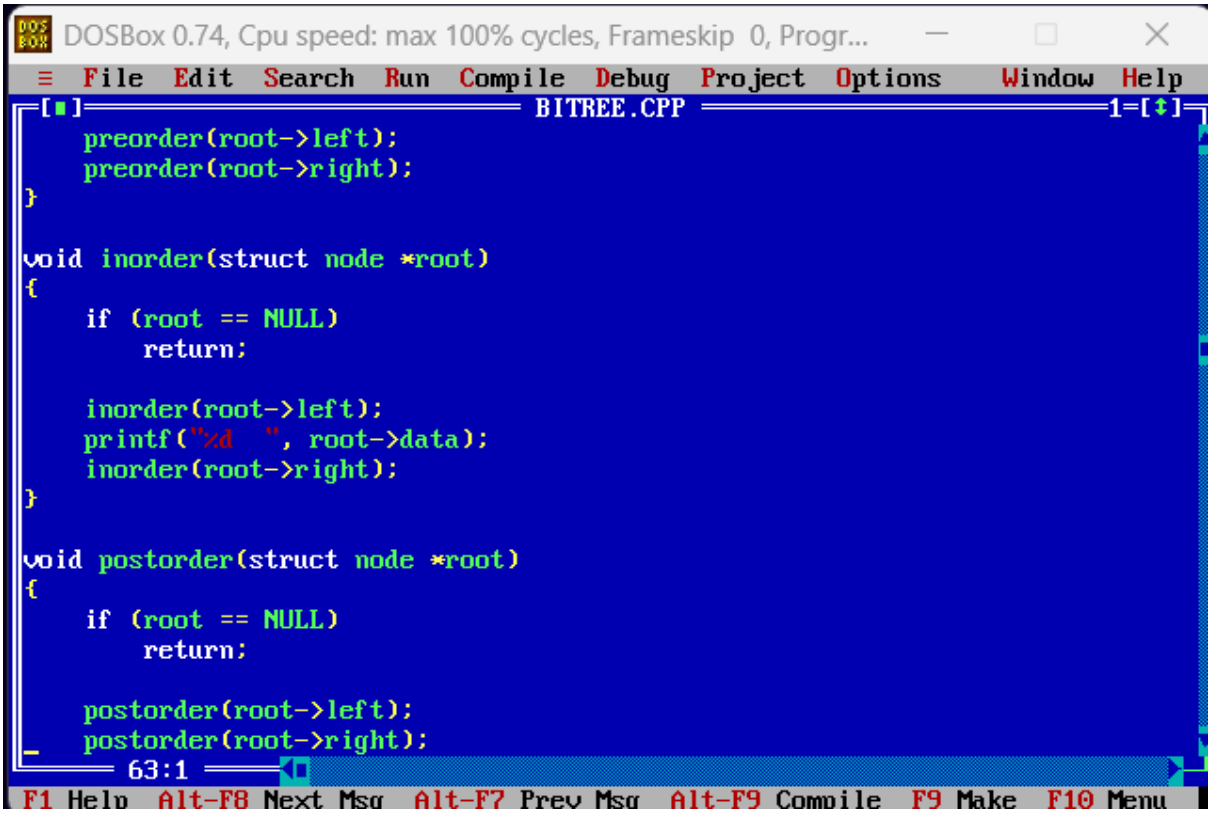
Name: Srushti Milap Patil
Roll no.: 66

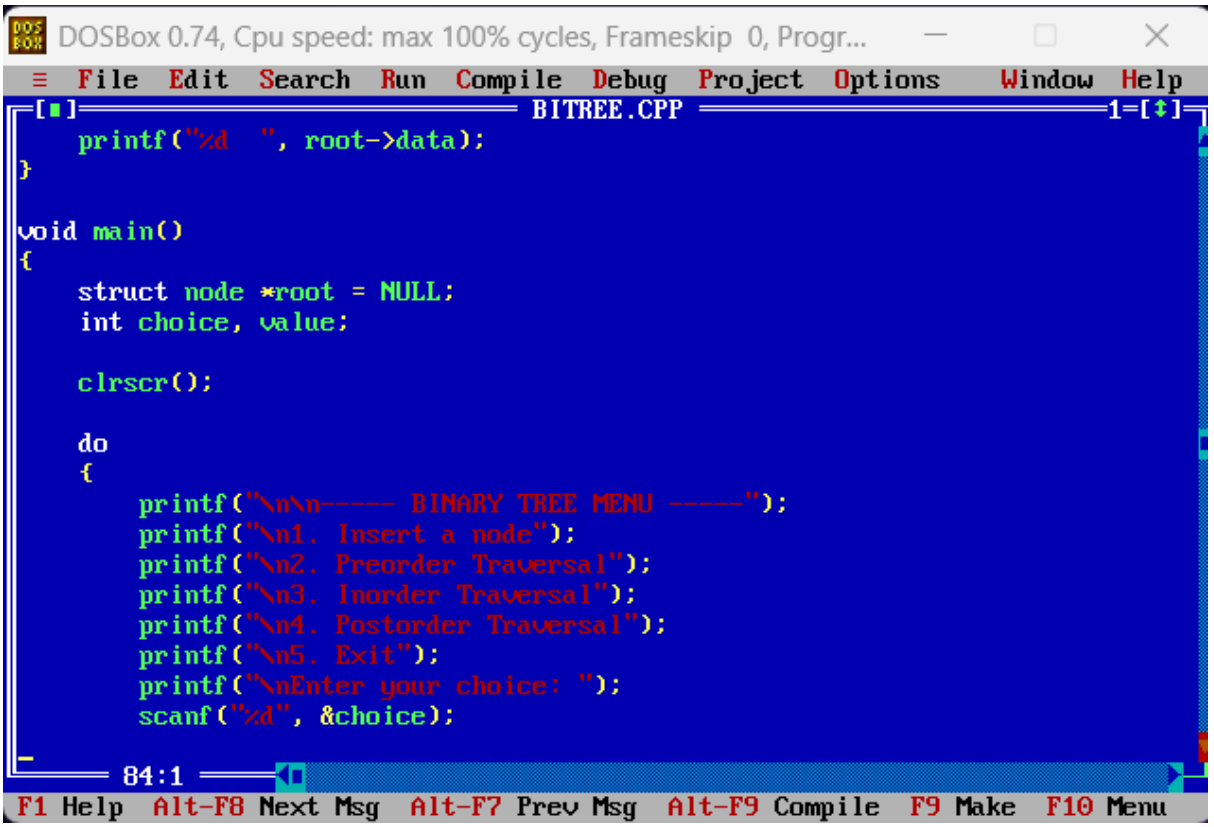
Div: BS4

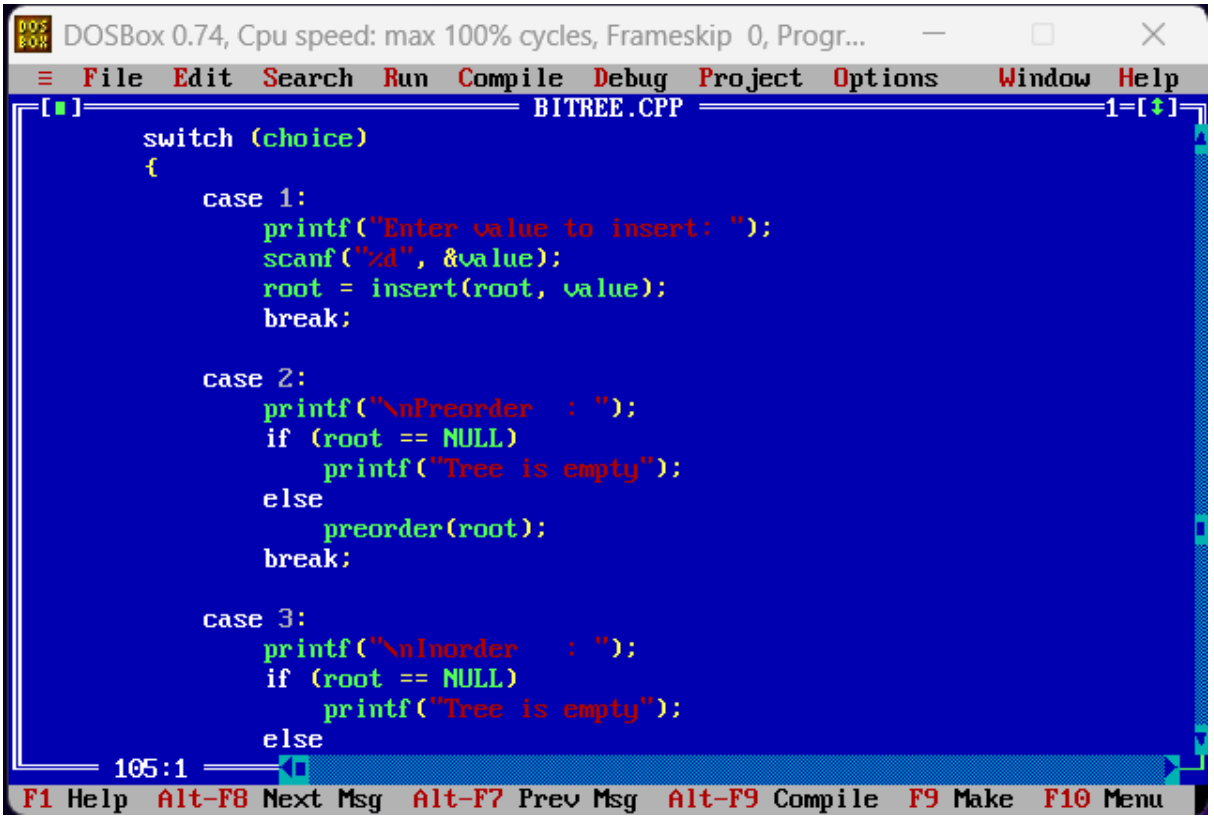
Program:

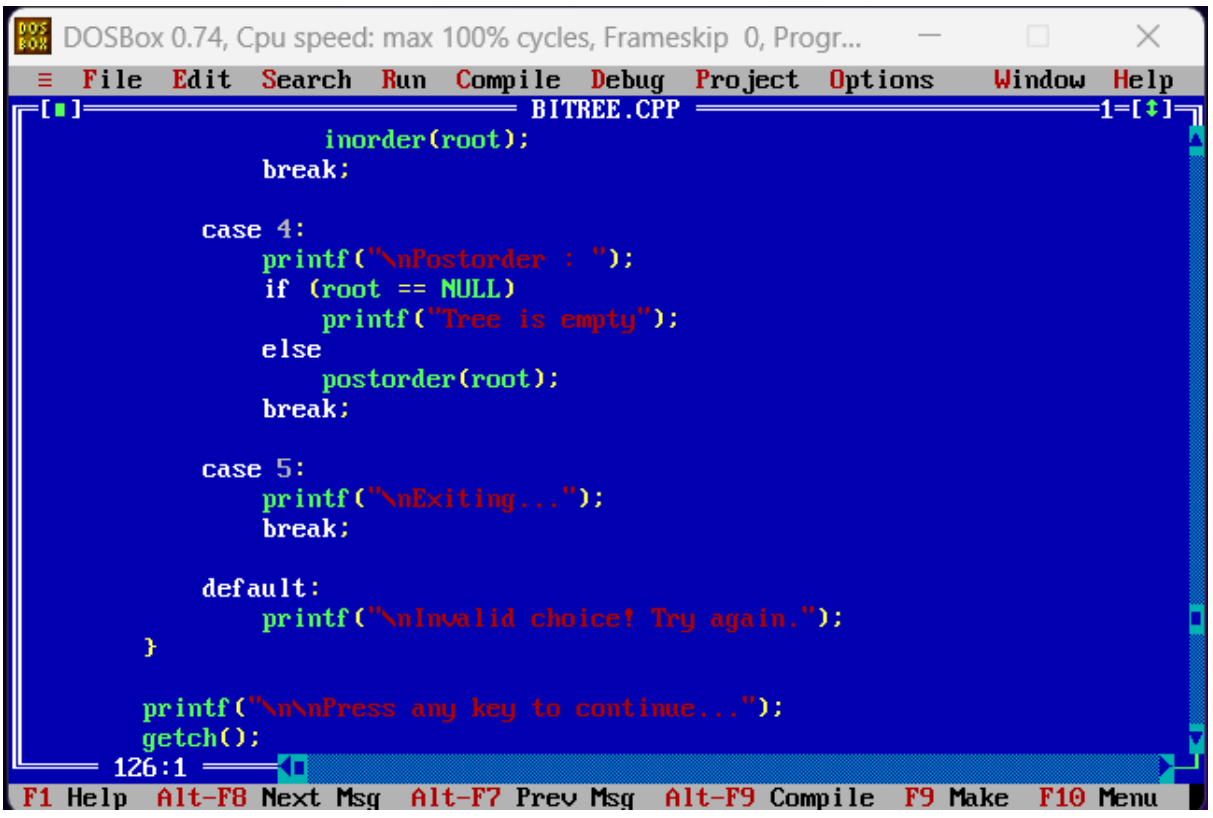


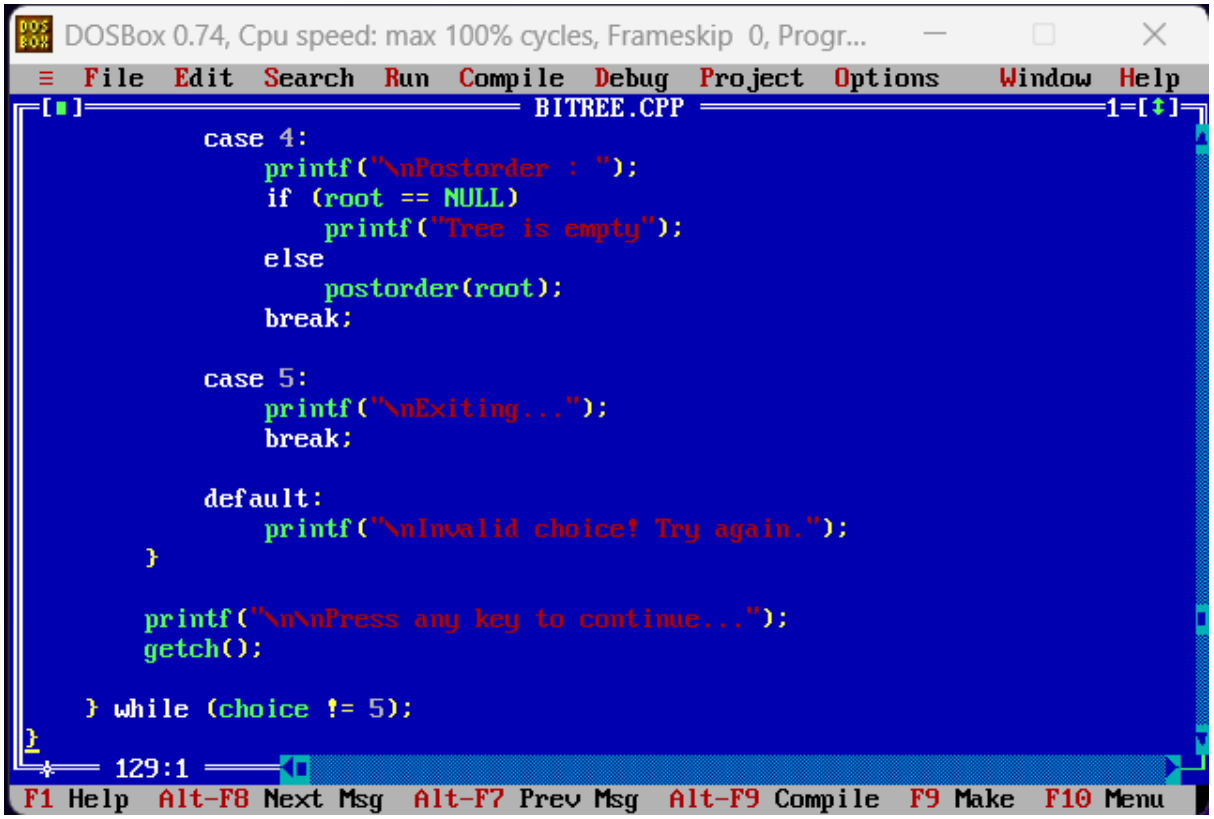



DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
BITREE.CPP
1
preorder(root->left);
preorder(root->right);
}
void inorder(struct node *root)
{
if (root == NULL)
return;
inorder(root->left);
printf("%d ", root->data);
inorder(root->right);
}
void postorder(struct node *root)
{
if (root == NULL)
return;
postorder(root->left);
postorder(root->right);
63:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

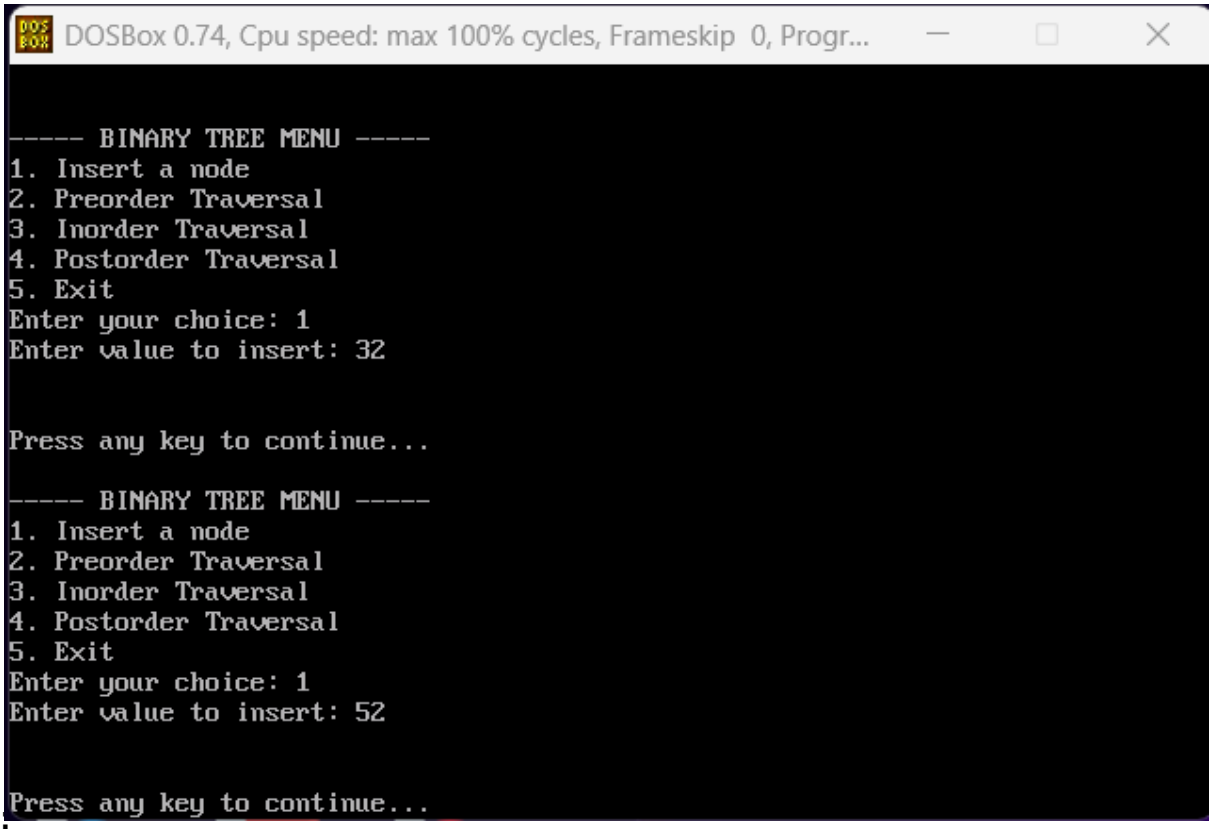

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
BITREE.CPP
1
printf("%d ", root->data);
}
void main()
{
struct node *root = NULL;
int choice, value;
clrscr();
do
{
printf("\n\n----- BINARY TREE MENU -----");
printf("\n1. Insert a node");
printf("\n2. Preorder Traversal");
printf("\n3. Inorder Traversal");
printf("\n4. Postorder Traversal");
printf("\n5. Exit");
printf("\nEnter your choice: ");
scanf("%d", &choice);
84:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
BITREE.CPP
switch (choice)
{
case 1:
printf("Enter value to insert: ");
scanf("%d", &value);
root = insert(root, value);
break;
case 2:
printf("\nPreorder : ");
if (root == NULL)
printf("Tree is empty");
else
preorder(root);
break;
case 3:
printf("\nInorder : ");
if (root == NULL)
printf("Tree is empty");
else
105:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

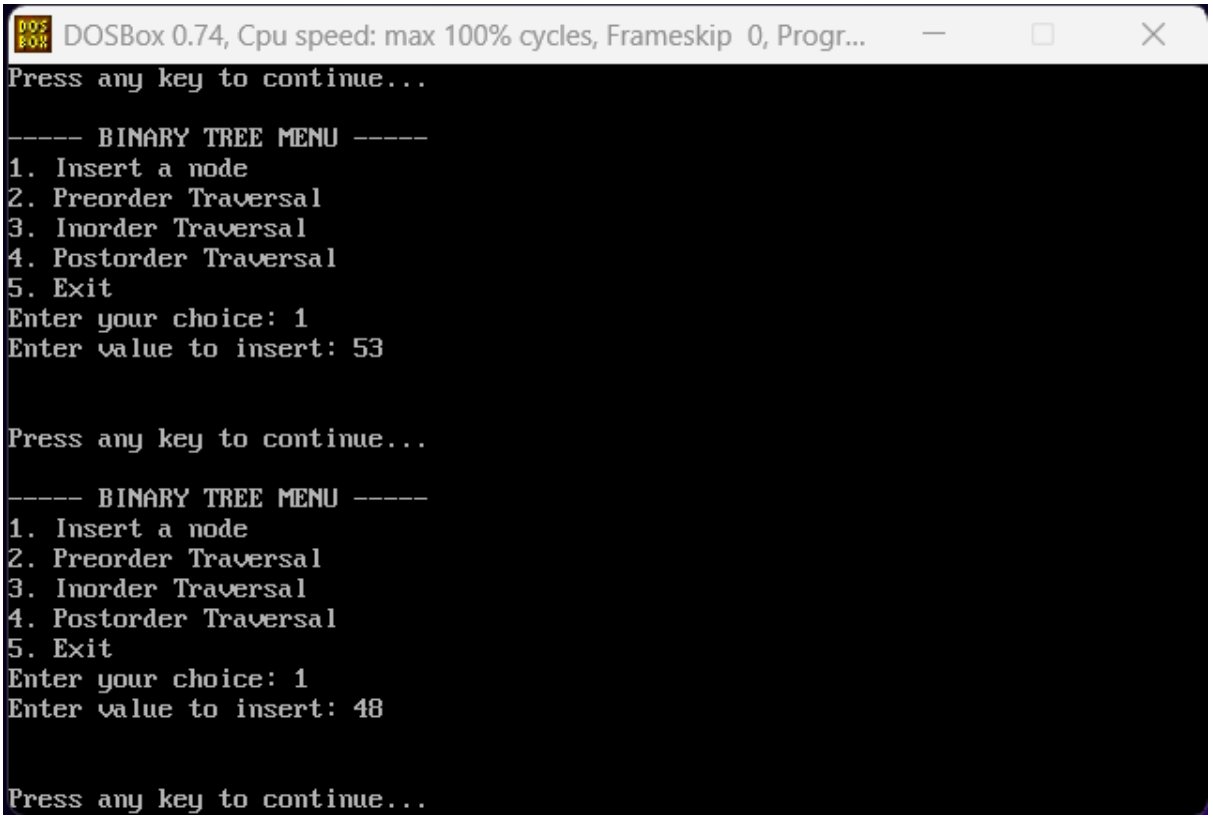

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
BITREE.CPP
inorder(root);
break;
case 4:
printf("\nPostorder : ");
if (root == NULL)
printf("Tree is empty");
else
postorder(root);
break;
case 5:
printf("\nExiting...");
break;
default:
printf("\nInvalid choice! Try again.");
}
printf("\n\nPress any key to continue...");
getch();
126:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

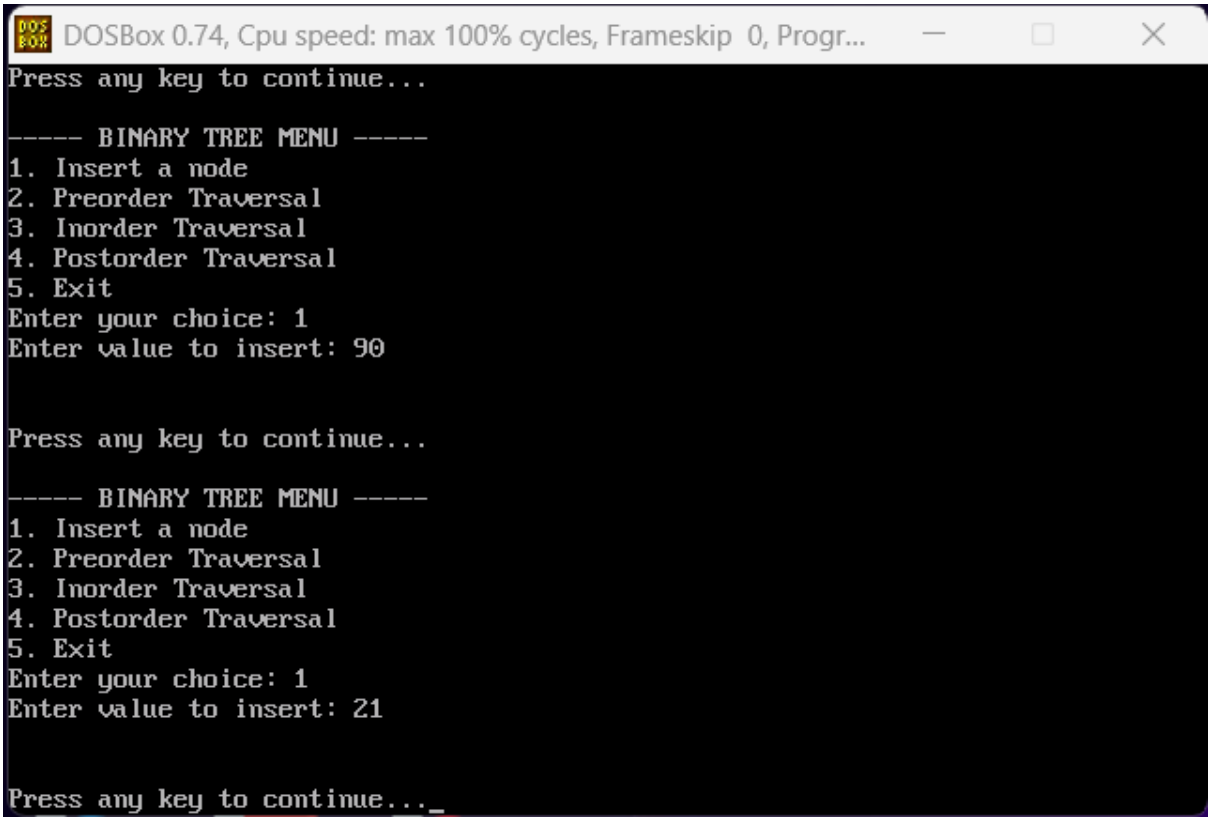

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
File Edit Search Run Compile Debug Project Options Window Help
BITREE.CPP
1
case 4:
printf("\nPostorder : ");
if (root == NULL)
printf("Tree is empty");
else
postorder(root);
break;
case 5:
printf("\nExiting...");
break;
default:
printf("\nInvalid choice! Try again.");
}
printf("\n\nPress any key to continue...");
getch();
} while (choice != 5);
}
129:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

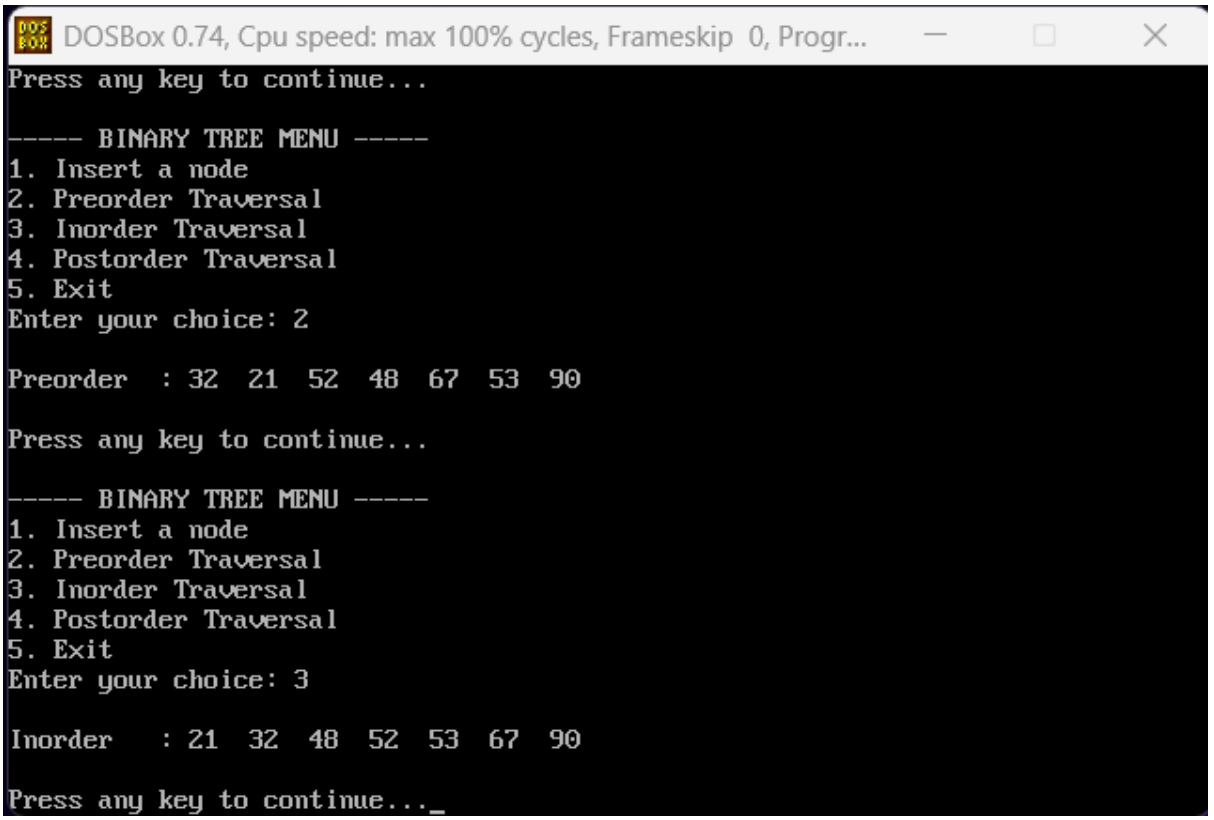
Output :

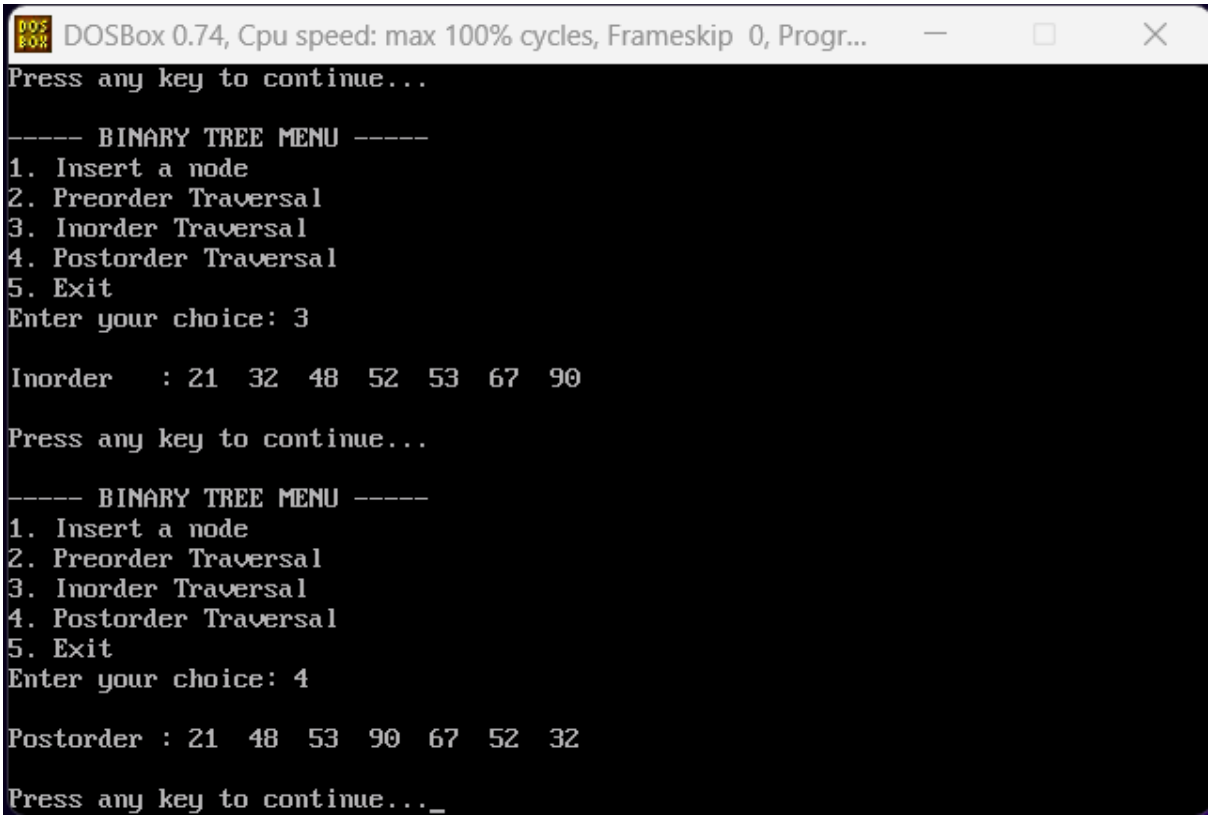

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 32
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 52
Press any key to continue...

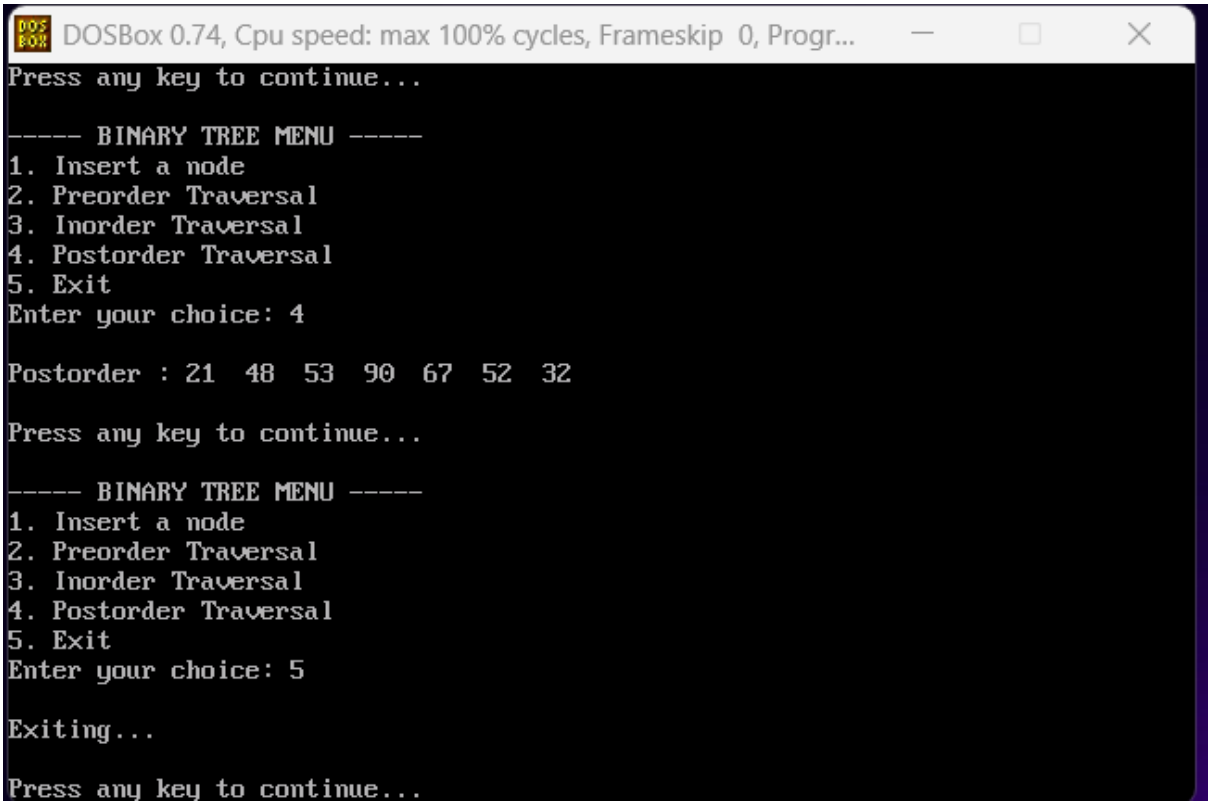

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 67
Press any key to continue...


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 53
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 48
Press any key to continue...


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 90
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 1
Enter value to insert: 21
Press any key to continue...


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 2
Preorder : 32 21 52 48 67 53 90
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 3
Inorder : 21 32 48 52 53 67 90
Press any key to continue...


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 3
Inorder : 21 32 48 52 53 67 90
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 4
Postorder : 21 48 53 90 67 52 32
Press any key to continue...


DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Progr...
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 4
Postorder : 21 48 53 90 67 52 32
Press any key to continue...
----- BINARY TREE MENU -----
1. Insert a node
2. Preorder Traversal
3. Inorder Traversal
4. Postorder Traversal
5. Exit
Enter your choice: 5
Exiting...
Press any key to continue...